

ECON108 Winter 2025

Lab Assignment 3

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One important thing to keep in mind is that percentage points and percentages are different, and you should use the correct unit! A change in a percentage is measured in percentage points.

Percentage points (%p or pp) $\neq$ Percentage

Example:

- The probability of getting a job with 3 interviews is 0.3 (or 30%).
- The probability of getting a job with 4 interviews is 0.35 (or 35%).
- If I currently have 3 interviews and get one more, my probability of getting a job increases by 5 **percentage points** (35% - 30% = **5pp**).

Question 1. What fraction of the sample support capital punishment for murder?

Answer: 58.55%

. tab cappun

Favor or oppose death penalty for murder	Freq.	Percent	Cum.
Not applicable	7,500	11.57	11.57
Favor	37,946	58.55	70.12
Oppose	15,604	24.08	94.19
Don't know	3,410	5.26	99.45
No answer	354	0.55	100.00
Total	64,814	100.00	

Question 2. Convert the following categorical variables into indicator variables (see the do file for guidance)

Please check the attached do-file.

Question 3. Use the sum command to compare the characteristics of supporters of capital punishment vs. non-supporters. What is different about these two populations?

```
.
. su postlife_y white black other_race male female educ age if cappun_fav == 1
```

Variable	Obs	Mean	Std. dev.	Min	Max
postlife_y	27,226	.8020642	.3984508	0	1
white	37,946	.860117	.3468702	0	1
black	37,946	.0923681	.2895487	0	1
other_race	37,946	.0475149	.2127403	0	1
male	37,946	.4787329	.4995541	0	1
female	37,946	.5212671	.4995541	0	1
educ	37,864	12.87579	2.965217	0	20
age	37,833	46.14524	17.37598	18	89

```
. su postlife_y white black other_race male female educ age if cappun_fav == 0
```

Variable	Obs	Mean	Std. dev.	Min	Max
postlife_y	11,516	.7911601	.4064974	0	1
white	15,604	.6825814	.4654868	0	1
black	15,604	.2446168	.4298735	0	1
other_race	15,604	.0728018	.2598193	0	1
male	15,604	.3717637	.4832912	0	1
female	15,604	.6282363	.4832912	0	1
educ	15,567	13.06816	3.500084	0	20
age	15,549	45.45122	17.87997	18	89

- The share of postlife believers, Whites and males is larger, while the share of Blacks, individuals of other races, and females is smaller among supporters of capital punishment compared to non-supporters. However, the average years of education and age are similar between the two groups, although the average age of supporters is slightly older than that of non-supporters.

Question 4. Estimate the following regression:

$$\text{cappunfor} = \beta_0 + \beta_1 \text{postlifeyes} + \beta_2 \text{sexmale} + \beta_3 \text{racewhite} + \beta_4 \text{educ} + \beta_5 \text{age} + u$$

Interpret the estimated coefficients.

```

. **Q4
. reg cappun_fav postlife_y male white educ age

```

Source	SS	df	MS	Number of obs	=	38,568
Model	420.494944	5	84.0989889	F(5, 38562)	=	424.89
Residual	7632.68572	38,562	.197932828	Prob > F	=	0.0000
				R-squared	=	0.0522
				Adj R-squared	=	0.0521
Total	8053.18067	38,567	.20881014	Root MSE	=	.4449

cappun_fav	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
postlife_y	.0222165	.0056722	3.92	0.000	.0110989	.0333341
male	.0875027	.0045794	19.11	0.000	.078527	.0964785
white	.2373934	.0057819	41.06	0.000	.2260608	.248726
educ	-.0067202	.0007461	-9.01	0.000	-.0081825	-.0052579
age	-.000425	.0001324	-3.21	0.001	-.0006845	-.0001654
_cons	.5626288	.0135072	41.65	0.000	.5361543	.5891033

- β_0 : The probability that a person supports capital punishment when they do not believe in an afterlife, are female, non-white, have zero years of education, and are newborn (age = 0) is 0.56 (56%).
(A newborn child, lol. Yes, this is yet another example of a meaningless y-intercept.)
- β_1 : The probability of supporting capital punishment is 2.22 percentage points (pp) higher for those who believe in an afterlife compared to those who do not, all else equal.
- β_2 : The probability of supporting capital punishment is 8.75pp higher for males compared to females, all else equal.
- β_3 : The probability of supporting capital punishment is 23.74pp higher for whites compared to non-whites, all else equal.
- β_4 : Each additional year of education decreases the probability of supporting capital punishment by 0.67pp, all else equal.
- β_5 : Each additional year of age decreases the probability of supporting capital punishment by 0.04pp, all else equal.
- All estimates are statistically significant at the 5% level.

Question 5. Predict the probability of supporting capital punishment for someone who believes in the afterlife, is female, is non-white, has 16 years of education, and is 40 years old.

Answer: 0.4603 (46.03%)

```
. **Q5
. margins, at(postlife_y=1 male = 0 white = 0 educ = 16 age = 40 )
```

Adjusted predictions
Model VCE: OLS

Number of obs = 38,568

Expression: Linear prediction, predict()

```
At: postlife_y = 1
    male       = 0
    white      = 0
    educ       = 16
    age        = 40
```

	Delta-method				
	Margin	std. err.	t	P> t	[95% conf. interval]
_cons	.460324	.0061526	74.82	0.000	.4482648 .4723833

Question 6. What is the change in probability of support for capital punishment if the person in Q5 receives one more year of education?

Answer: -.0067 (-0.67pp)

```
. margins, dydx(educ) at(postlife_y=1 male = 0 white = 0 age = 40 )
```

Average marginal effects
Model VCE: OLS

Number of obs = 38,568

Expression: Linear prediction, predict()

dy/dx wrt: educ

At: postlife_y = 1
male = 0
white = 0
age = 40

	Delta-method					
	dy/dx	std. err.	t	P> t	[95% conf. interval]	
educ	-.0067202	.0007461	-9.01	0.000	-.0081825	-.0052579

Question 7. Generate the predicted probability support for capital punishment for each observation in the dataset. How many predictions are out of the $[0,1]$ range?

All predictions fall within the $[0, 1]$ range.

```
. **Q7
. predict phat_lpm, xb
(22,450 missing values generated)
```

```
. su phat_lpm
```

Variable	Obs	Mean	Std. dev.	Min	Max
phat_lpm	42,364	.7023202	.1047704	.3916789	.8923183

Question 8. What is the count R-squared? Do you think this model does a good job at predicting the probability of support for capital punishment?

Answer: 0.6411 “Yes, 64.11% of the sample was correctly predicted.”

```
.
. **Q8
. gen correct_lpm = 0

. replace correct_lpm = 1 if phat_lpm < 0.5 & cappun_fav == 0
(1,936 real changes made)

. replace correct_lpm = 1 if phat_lpm > 0.5 & cappun_fav == 1
(36,058 real changes made)

. replace correct_lpm = . if phat_lpm == .
(22,450 real changes made, 22,450 to missing)

.
. su correct_lpm
```

Variable	Obs	Mean	Std. dev.	Min	Max
correct_lpm	42,364	.6411104	.4796804	0	1

Question 9. Instead of a linear probability model, estimate support for capital punishment using a probit regression:

$$P(\text{cappunfor} = 1) = \Phi(\beta_0 + \beta_1 \cdot \text{postlife_y} + \beta_2 \cdot \text{sexmale} + \beta_3 \cdot \text{racewhite} + \beta_4 \cdot \text{educ} + \beta_5 \cdot \text{age})$$

Interpret the estimated coefficients.

```
. **Q9
. probit cappun_fav postlife_y male white educ age

Iteration 0:  Log likelihood = -23462.074
Iteration 1:  Log likelihood = -22489.975
Iteration 2:  Log likelihood = -22488.987
Iteration 3:  Log likelihood = -22488.987

Probit regression                               Number of obs =  38,568
                                                LR chi2(5)      = 1946.17
                                                Prob > chi2     =  0.0000
Log likelihood = -22488.987                    Pseudo R2      =  0.0415
```

cappun_fav	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
postlife_y	.0682034	.017037	4.00	0.000	.0348115	.1015954
male	.2675692	.0139475	19.18	0.000	.2402326	.2949059
white	.6510269	.0166601	39.08	0.000	.6183737	.6836801
educ	-.0200533	.002216	-9.05	0.000	-.0243966	-.01571
age	-.0012538	.0003989	-3.14	0.002	-.0020356	-.000472
_cons	.1752168	.0399807	4.38	0.000	.0968561	.2535775

- β_0 : The z-score of a person who does not believe in an afterlife, is female, non-white, has zero years of education, and is zero years old is 0.175.
- β_1 : The z-score for someone who believes in an afterlife is 0.068 higher than for a person who does not believe in an afterlife, all else equal.
- β_2 : The z-score for males is 0.268 higher than for females, all else equal.
- β_3 : The z-score for whites is 0.651 higher than for non-whites, all else equal.
- β_4 : For each additional year of education, the z-score decreases by 0.02, all else equal.
- β_5 : For each additional year of age, the z-score decreases by 0.001, all else equal.
- All estimates are statistically significant at the 5% level.

Question 10. Using your probit model estimates, predict the probability of supporting capital punishment for someone who believes in the afterlife, is female, is non-white, has 16 years of education, and is 40 years old.

Answer: 0.449 (44.9%)

```
. **Q10
. margins, at(postlife_y=1 male = 0 white = 0 educ = 16 age = 40 )

Adjusted predictions                                Number of obs = 38,568
Model VCE: OIM

Expression: Pr(cappun_fav), predict()
At: postlife_y = 1
    male      = 0
    white     = 0
    educ      = 16
    age       = 40
```

	Delta-method				
	Margin	std. err.	z	P> z	[95% conf. interval]
_cons	.4492389	.0070147	64.04	0.000	.4354902 .4629875

Wait, what is the difference between the `margins` and `predict` commands?

- **margins** calculates **average predicted values for groups of observations**. For example, the average predicted values for non-white, 40-year-old women with 16 years of education. Stata presents the results as a table.
- **predict** calculates **a predicted value for each observation** in the data, using the values of the explanatory variables for that observation. For example, it generates a predicted value for every individual in the dataset. To store these values, we create a new variable (e.g., `yhat`). To check the predicted values, we browse the dataset.

Question 11. What is the change in probability of support for capital punishment if the person in Q10 receives one more year of education?

Answer: -.008 (-0.8pp)

```
. *Q11
. margins, dydx(educ) at(postlife_y=1 male = 0 white = 0 age = 40 )

Average marginal effects                                Number of obs = 38,568
Model VCE: OIM

Expression: Pr(cappun_fav), predict()
dy/dx wrt:  educ
At: postlife_y = 1
    male      = 0
    white     = 0
    age       = 40
```

	Delta-method		z	P> z	[95% conf. interval]	
	dy/dx	std. err.				
educ	-.0079662	.0008763	-9.09	0.000	-.0096838	-.0062487

Comment: I realized that the correct command for the probit model should include `educ = 16` inside `at()` because the model is nonlinear. When I add this, the estimated slope is -0.0079353, which is very close to -0.0079662. At age 12, the estimated effect is -0.0079912. Since the slope remains nearly constant across all age levels, this suggests an approximately linear relationship between age and the outcome variable.

Question 12. Generate the predicted probabilities (call it phat_probit) and calculate the count R-squared.

Answer: 0.6193 “61.93% of the sample was correctly predicted.”

```
. *Q12
. predict phat_probit, xb
(22450 missing values generated)

. su phat_probit
```

Variable	Obs	Mean	Std. dev.	Min	Max
phat_probit	42,364	.5510599	.2940306	-.3336754	1.110611

```
.
.
. gen correct_probit = 0

. replace correct_probit = 1 if phat_probit < 0.5 & cappun_fav == 0
(4,734 real changes made)

. replace correct_probit = 1 if phat_probit > 0.5 & cappun_fav == 1
(32,334 real changes made)

. replace correct_probit = . if phat_probit == .
(22,450 real changes made, 22,450 to missing)

.
. su correct_probit
```

Variable	Obs	Mean	Std. dev.	Min	Max
correct_probit	42,364	.6192522	.4855764	0	1

Question 13. Which model would you prefer? Why?

I prefer the linear probability model (LPM) because all predicted values lie within the range [0,1], and the count R-squared is higher than that of the probit model, indicating that the LPM has better predictability.